DID EPRC Workshop Talk

Fleshed-Out Slide Outline for Parts A–C

Draft working document

October 2026

Purpose

This document turns the rough workshop plan into a slide-by-slide outline. The talk is designed as a 45-minute introduction to difference-in-differences (DID) methods for education policy researchers. The first pass focuses on:

- Part A: a traditional DID example using simulated education-policy data.
- Part B: the regression version, two-way fixed effects (TWFE), and an event-study extension.
- Part C: a light transition to staggered DID, leaving most details for Wei's section.

Running Example

The motivating education-policy example is entry into a gifted and talented (G&T) program:

> Do students, schools, or districts exposed to a G&T policy change experience larger achievement gains than a comparable group not yet exposed to the policy?

For the first workshop version, the data are simulated district-year panel data. The story can be translated into student-level or school-level language later, but the simplest teaching version should keep the design at one level: one unit, one time variable, one treatment timing variable, and one outcome.

Data Used in the Examples

File	Use in the talk
`simple_did_district.csv/.dta`	Main teaching data for traditional DID. Half of units are treated starting in year 5; half are never treated.
`staggered_did_district.csv/.dta`	Preview data for staggered DID. Some units adopt in year 5, some adopt in year 8, and some are never treated.
`workshop_estimates.csv`	Small reference table comparing estimated effects to the known simulated treatment effect.

Slide Map

Slide	Section	Main point	Suggested visual

1	Opening	Title slide and workshop goals.	Title plus 3 learning goals.
2	Traditional DID	One education-policy example of traditional DID and the idea of parallel trends.	`simple_parallel_trends.png`.
3	Data setup	What data do we need for DID? Units, time, outcome, treatment, comparison group.	Variable/design table.
4	DID contrast	DID compares changes, not only levels.	`simple_did_2x2.png`.
5	Regression version	The two-by-two contrast can be written as an interaction model.	Equation plus estimate/truth table.
6	TWFE	Add unit and year fixed effects to absorb stable unit differences and common shocks.	TWFE equation and interpretation.
7	Event study	Replace one post-treatment indicator with event-time indicators.	`simple_event_study.png`.
8	Limitation	A single TWFE coefficient can hide dynamic treatment effects.	Return to event-study figure.
9	Staggered transition	What changes when treatment begins at different times?	`staggered_timing_plot.png`.
10	Staggered preview	Early and later treatment groups can both be treated, but at different event times.	`staggered_group_trends.png`.
11	Handoff	Preview Wei's section: TWFE problems, CSA, ETWFE, covariates, clustering, PT/CPT.	Roadmap slide.
12	Lab setup	Orient students to the website, data, and code.	Workshop website links.

Detailed Slide Outline

Slide 1. Title Slide

Purpose. Set up the session as a practical 45-minute introduction to DID for education policy.

Draft slide title. Difference-in-Differences for Education Policy.

On-slide bullets.

- What DID estimates.
- What parallel trends means.
- How the same idea appears in regression, TWFE, and event-study form.
- Why staggered treatment timing requires extra care.

Speaker note. Emphasize that the examples use generated data so the true treatment effect is known.

Slide 2. Traditional DID Example and Parallel Trends

Purpose. Start with intuition: treated and comparison units are followed before and after a policy change.

Example language. Suppose some districts adopt a G&T access policy in year 5 and other districts do not. DID asks whether achievement changes more in the treated districts than in the comparison districts after adoption.

Main point. Parallel trends is the claim that, absent treatment, the treated group would have continued to move like the comparison group.

Figure placeholder: `simple_parallel_trends.png`
Traditional DID example: one treated group adopts in year 5 and one comparison group is never treated.

Slide 3. What Data Do We Need?

Purpose. Make the required panel structure concrete before notation.

DID ingredient	G&T example	Simulated variable
Unit	Student, school, or district	`district_id`
Time	Year or grade	`year`
Outcome	Achievement	`outcome`
Treatment group	Units exposed to policy	`treated_ever`
Post period	Years after adoption	`post`
Treatment timing	First exposure year	`treatment_start_year`

Speaker note. The current generated data are district-year, but the same logic applies to student-year or school-year panels.

Slide 4. DID Compares Changes

Purpose. Show the two-by-two DID contrast visually.

Main point. DID is not simply treated versus comparison after the policy. It is the treated group's change minus the comparison group's change.

$$\widehat{\delta}_{DID} = (\bar{Y}_{T,post} - \bar{Y}_{T,pre}) - (\bar{Y}_{C,post} - \bar{Y}_{C,pre})$$

Figure placeholder: `simple_did_2x2.png`
The two-by-two DID contrast compares changes from pre to post across groups.

Slide 5. The Simple DID Regression

Purpose. Connect the visual contrast to the regression coefficient.

$$Y_{it} = \alpha + \beta_1 Treat_i + \beta_2 Post_t + \delta(Treat_i \times Post_t) + \varepsilon_{it}$$

Model	Estimated effect	Known simulated effect
Two-by-two DID regression	0.087	0.090

Speaker note. This slide is where we say why simulation is helpful: students can see that the estimator is recovering something close to the known effect.

Slide 6. Two-Way Fixed Effects

Purpose. Move from the simplest DID regression to the common TWFE form.

$$Y_{it} = \alpha_i + \lambda_t + \delta D_{it} + \varepsilon_{it}$$

On-slide bullets.

- Unit fixed effects, α_i, absorb stable differences across units.
- Year fixed effects, λ_t, absorb shocks common to all units in a given year.
- D_{it} turns on once the unit is treated.
- In the simple one-cohort case, TWFE lines up with the basic DID logic.

Speaker note. The goal is not to over-teach TWFE here; it is to show that the familiar DID contrast is the same idea students see in applied papers.

Slide 7. Event-Study Version

Purpose. Show how the design becomes a set of event-time coefficients.

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k 1[t - G_i = k] + \varepsilon_{it}$$

Main point. Event studies help us look at pre-treatment differences and dynamic post-treatment effects.

Figure placeholder: `simple_event_study.png`
Event-study coefficients for the simple DID example, using year -1 as the reference period.

Slide 8. What TWFE Can Hide

Purpose. Introduce the first limitation without getting too technical.

On-slide bullets.

- The TWFE coefficient gives one average effect.
- In the simulated data, the treatment effect grows over time.
- A single coefficient can be useful, but it can hide dynamics.
- Event-study plots make those dynamics more visible.

Speaker note. This slide sets up the next question: if one treatment date already requires care, what happens when treatment timing differs across groups?

Slide 9. Transition to Staggered DID

Purpose. Introduce multiple treatment timings as the bridge to Wei's section.

Example language. In the G&T example, students may enter at different grade points. In the simulated data, one cohort is treated in year 5 and another cohort is treated in year 8.

Figure placeholder: `staggered_timing_plot.png`
Staggered preview: early adopters, later adopters, and never-treated units.

Slide 10. Staggered Group Trends

Purpose. Show that the same policy can create multiple treated groups over time.

On-slide bullets.

- Early adopters are treated beginning in year 5.
- Later adopters are untreated until year 8.
- Never-treated units are untreated throughout.
- The comparison group can change depending on the estimand and method.

Figure placeholder: `staggered_group_trends.png`
Mean trends for early-treated, later-treated, and never-treated units.

Slide 11. Handoff to Wei's Staggered DID Section

Purpose. Mark the boundary between the introductory section and the more advanced staggered DID section.

Placeholder bullets for Wei.

- Why TWFE can be problematic with staggered timing and heterogeneous effects.
- Goodman-Bacon decomposition intuition.
- Callaway–Sant'Anna: estimate group-time ATT(g, t) and aggregate.
- Wooldridge pooled OLS / ETWFE as another route.
- Covariates, conditional parallel trends, clustering, and inference.

Slide 12. Lab and Materials

Purpose. Point students to the workshop site and files.

On-slide bullets.

- Slides/outline: downloadable PDF from the workshop website.

- Data: CSV and Stata versions of the simulated data.
- Code: simple DID, event study, and staggered preview scripts.
- Resources: citations for follow-up reading.

Website placeholder. https://katjanestrickland.github.io/did-eprc-workshop/